```
clc;
clear;
close all;
```

1. SYNTHETIC SPEECH GENERATION

```
Fs = 16000;                  % Sampling rate (Hz)
duration = 0.5;               % Duration (seconds)
t = 0:1/Fs:duration-1/Fs;

f1 = 250;
f2 = 700;

x_speech = sin(2*pi*f1*t) + 0.5*sin(2*pi*f2*t);
```

2. FRAME-BY-FRAME LPC VOCODER SYNTHESIS

```
p = 12;                       % Prediction order
N_frame = 320;
hop = 160;

num_frames = floor((length(t)-N_frame)/hop) + 1;
x_synth = zeros(1, length(t));

rng(42);                      % Fixed random seed for noise excitation

for m = 1:num_frames
    idx = (m-1)*hop + (1:N_frame);

    frame = x_speech(idx).*hamming(N_frame)';

    % Compute LPC coefficients using Levinson-Durbin
    [a, g] = lpc(frame, p);

    % Excitation generation and IIR synthesis filtering
    excitation = sqrt(g)*randn(1, N_frame);
    synth_frame = filter(1, a, excitation);

    x_synth(idx) = x_synth(idx) + synth_frame;
end
```

3. SINGLE FRAME SPECTRAL ANALYSIS

```
frame_sample = x_speech(161:160+N_frame).*hamming(N_frame)';
[a_sample, g_sample] = lpc(frame_sample, p);

[H_lpc, w] = freqz(sqrt(g_sample), a_sample, 1024, Fs);
```

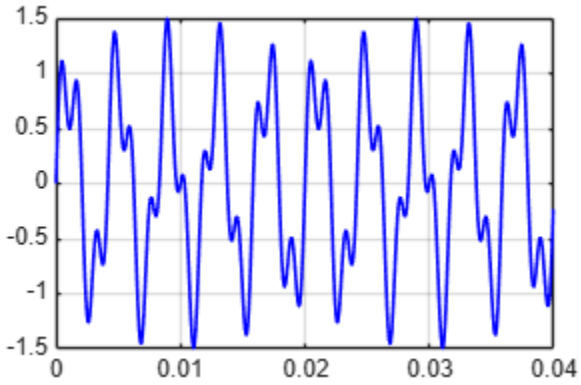
```
[X_fft, f_fft] = pwelch( ...
    frame_sample, hamming(N_frame), [], N_frame, Fs);
```

4. SUBFIGURE PLOTTING AND EXPORT

```
figW = 8.5;
figH = 5.5;

% Subfigure 20A: Original Speech Signal
figA = figure('Units', 'centimeters', ...
    'Position', [2, 2, figW, figH]);

plot(t(1:640), x_speech(1:640), 'b', 'LineWidth', 1.2);
grid on;
box on;
```



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